PROJECT ARCHITECTURE & CAPABILITIES DECK

Blus Invoice Agent

Enterprise Accounts Payable Duplicate & Fraud Detection Engine

1. Multi-Format Document Extraction (PDF Text Layer, Scanned PDF OCR, Images, QR JWT, CSV & Excel)
2. Deterministic 10-Step Verification Pipeline & Fraud Rules Engine (AP-02, AP-03, AP-09, AP-11)
3. High-Volume Background Job Queue (Up to 2,000 Invoices/Batch with Pause, Cancel & Resume)
4. Live Read-Only SAP S/4HANA Cloud OData V2 Sandbox Integration
5. Grounded Read-Only AI Chat Assistant with 17-Intent Router & 6 Live Ledger Lookup Tools
6. Tamper-Evident SHA-256 Audit Log Chain, PII Redaction Guard & SAP BTP MTA Deployment

`Runtime Stack: Node.js 22 | SAP CAP (@sap/cds) | Express | SQLite / SAP HANA HDI | pdfjs-dist | Tesseract.js | Google Gemini SDK`

1. Project Goal & Business Problem Solved

Why Blus Invoice Agent exists and what it protects against

The Accounts Payable Problem

RISK

- Suppliers re-submit identical invoices via email, PDF, and EDI.
- Fraudsters or clerical errors alter dates/totals on duplicate invoice numbers.
- High-value purchases are split into multiple smaller bills just below approval thresholds (e.g. INR 49,500 x 3) to bypass manager sign-off.
- Multiple vendor master IDs share the same GSTIN to hide related-party billing.
- Manual review of 1,000+ Fiori CSV/Excel rows or scanned PDFs is slow and error-prone.

The Blus Invoice Agent Goal

SOLUTION

- Catch duplicate, repeat, and split invoices BEFORE they are posted or paid.
- Provide 100% deterministic rule verdicts: PASS, HOLD, or REJECT.
- Automatically save clean (PASS) invoices into the central Ledger for future cross-checks.
- Route suspicious (HOLD/REJECT) bills to a dedicated Controller Review Queue.
- Give AP teams a fast, natural-language Chat Assistant to query the ledger, review queue, and check reasons without risking unauthorized edits.

Core Design Philosophy: Deterministic Rules + Read-Only AI

ARCHITECTURE PRINCIPLE

Deterministic Verdicts (Never AI-Guessed)

- Every PASS, HOLD, or REJECT decision is computed strictly by deterministic code in srv/rules/rules.js and srv/lib/extract.js. The LLM never decides whether an invoice passes or fails.

Strictly Read-Only AI Assistant

- The Chat Assistant can read check results, search the ledger, and explain why a check failed, but has zero write permissions (cannot approve, skip checks, post, or delete records).

Privacy-First Grounding

- By default (ALLOW_INVOICE_DATA_TO_LLM=false), supplier names, GSTINs, bank accounts, invoice numbers, and amounts are redacted before any follow-up question reaches Gemini.

2. End-to-End System Architecture

Four-layer modular architecture built on SAP CAP (Node.js) and Express

1. Web UI Layer (app/)

FRONTEND

- app/index.html & app/app.js
- 6 Unified Workspace Views:
- 1. Upload invoice (Chat + File chip)
- 2. S/4HANA extract (Batch & SAP)
- 3. Ledger (Saved PASS invoices)
- 4. Review queue (HOLD/REJECT)
- 5. Check history (All checks)

2. Service Layer (server.js

CAP + EXPRESS

- server.js + srv/fraud-service.cds/.js
- OData V4 actions & Express REST routes
- Role-Based Access Control (XSUAA / Basic / Role headers)
- Streaming CSV & Excel (.xlsx) exports
- Server-side pagination (50 rows/page)
- UTF-8 JSON response sanitization

3. Core Engines (srv/lib/ & rules

ENGINES

- pdftext.js / pdfimage.js / ocr.js
- extract.js (Field parser & doc classifier)
- rules/rules.js (14 AP fraud rules)
- queue.js (Chunked 2000-row batch jobs)
- sap.js (SAP S/4HANA OData V2 client)
- agent-tools.js (17-intent router & 6 tools)

4. Database Entities (db/schema.cds) & Cloud Foundry Deployment (mta.yaml)

DATA & CLOUD

Core Database Tables (SQLite In-Memory for Dev/Preview | SAP HANA HDI Container in Production)

- ap.Posted (Ledger of verified invoices, indexed by supplierGstin + fiscalYear + invoiceNo) | ap.Checks (Every check run + JSON snapshot) | ap.CheckResults (Rule-level outcomes) | ap.Alerts (Open HOLD/REJECT items for Review Queue) | ap.AuditLog (SHA-256 hash chain) | ap.Jobs (Batch job state)

SAP BTP Multi-Target Application (mta.yaml)

- Three modules: blus-invoice-agent-srv (Node.js service), blus-invoice-agent-db-deployer (@sap/hdi-deploy), and blus-invoice-agent-approuter (@sap/approuter + XSUAA).

3. What We Used for PDF, Image & File Extraction

Multi-stage extraction pipeline in srv/lib/ handling digital PDFs, scanned PDFs, images, QR codes & spreadsheets

Stage A: Digital PDFs & Scanned PDFs

PDF PIPELINE

1. pdfjs-dist (srv/lib/pdftext.js)

- Extracts embedded text directly from digital PDF pages using Mozilla PDF.js (pdfjs-dist/legacy/build/pdf.mjs) without needing OCR.

2. @napi-rs/canvas (srv/lib/pdfimage.js)

- Renders PDF pages into high-resolution PNG image buffers in memory when a PDF has no text layer (scanned PDF) or contains an e-Invoice QR code.

3. tesseract.js + eng.traineddata (srv/lib/ocr.js)

- Runs local Optical Character Recognition (OCR) on rendered PDF page images and

Stage B: QR Codes, Spreadsheets & AI Fallback

QR / CSV / AI

4. jsqr + Node crypto (e-Invoice QR JWT)

- Decodes QR codes from invoice images and verifies signed Indian e-Invoice JWTs (RS256) against the IRP public key.

5. Custom Fiori CSV Parser & ExcelJS (srv/lib/csv.js)

- Maps SAP S/4HANA Fiori export columns (SupplierInvoice, InvoicingParty, InvoiceGrossAmount, PostingDate) and .xlsx sheets.

6. Opt-In Gemini AI Assist (srv/lib/ai-extract.js)

- If a noisy scan/photo fails standard extraction (COULD NOT READ), the user can click

Stage C: Structured Field Extraction & Document Type Validation (srv/lib/extract.js)

FIELD PARSER

11 Extracted Invoice Fields

- Supplier Name | Supplier GSTIN (15-char) | Buyer GSTIN | Invoice Number | Invoice Date | PO Number | Taxable Amount | CGST / SGST / IGST | Total Gross Amount | Bank IFSC & Account Number | Line Description.

Non-Invoice & Incomplete Guards

- Automatically detects and stops non-invoice files (Purchase Orders, Bank Statements, Delivery Challans, Quotes -> NOT AN INVOICE) and flags invoices missing Invoice No, Date, Total, or Supplier Identity (-> INCOMPLETE).

4. How the 10-Step Invoice Check Pipeline Works

Every uploaded invoice executes a transparent, auditable 10-step verification sequence

Steps 1 to 5: Ingestion, Reading & Data Quality

STEPS 1-5

Step 1: Receive & Validate File

- Validates file extension (.pdf, .png, .jpg, .jpeg, .csv, .xlsx) and enforces the 10 MB maximum file size limit.

Step 2: Detect Document Format

- Identifies PDF with text layer, Scanned PDF, Image (PNG/JPG), Excel (.xlsx), or CSV.

Step 3: Read the Document

- Runs pdfjs-dist text extraction, falls back to @napi-rs/canvas + Tesseract OCR if text < 20 chars, and decodes any QR JWT.

Step 4: Extract Fields & Classify Document

- Parses header, tax, total, and bank fields; checks if document is a valid supplier invoice and whether mandatory fields exist.

Step 5: Check Arithmetic & GSTIN Format

- Verifies Taxable + CGST + SGST + IGST == Total (within INR 1.00 tolerance) and validates the 15-character GSTIN format & Modulo-36 check digit.

Steps 6 to 10: Rules, Lookups, Ledger & Audit

STEPS 6-10

Step 6: Run Active Fraud Rules (AP-02, AP-03, AP-09, AP-11)

- Evaluates the 4 active rules against the Ledger (ap.Posted) and Vendor Master (data/masters.json).

Step 7: Look Up Similar Past Invoices & SAP Match

- Queries indexed ledger records by Supplier GSTIN + Fiscal Year and checks live SAP S/4HANA sandbox if configured.

Step 8: Compute Deterministic Verdict

- Assigns PASS, HOLD, REJECT, INCOMPLETE, NOT AN INVOICE, or COULD NOT READ.

Step 9: Persist to Check Log & Hash-Chained Audit Log

- Saves record to ap.Checks, creates Review Queue alert in ap.Alerts if HOLD/REJECT, and appends SHA-256 hash to ap.AuditLog.

Step 10: Auto-Save PASS to Ledger & Render Result

- Automatically inserts PASS invoices into ap.Posted so any future duplicate upload is immediately caught as REJECT.

5. Fraud & Duplicate Detection Rules Engine

4 Active Core Rules by default + 10 configurable enterprise AP rules in srv/rules/rules.js

Duplicate Check (AP-02) -> REJECT

ACTIVE RULE 1

- Normalizes Supplier GSTIN, Fiscal Year (April-March), and Invoice Number (stripping spaces/punctuation) or IRN.
- If exact same date & total exist in ledger: "This invoice is already saved in the ledger. Uploading it again is treated as a duplicate."
- If date or total differs: "Duplicate of invoice <No> from the same supplier, same financial year (<FY>), with a different date or total."

GSTIN Check (AP-03) -> REJECT

ACTIVE RULE 2

- Cross-checks the invoice Supplier GSTIN against all vendor records in data/masters.json.
- Flags REJECT if a single GSTIN is registered to more than one distinct Vendor ID (unless listed in the parent/affiliate exemption table).
- Prevents shell vendor creation and duplicate vendor master payouts.

Repeat Bill Check (AP-09) -> REJECT

ACTIVE RULE 3

- Detects when the same supplier submits another invoice with a different invoice number for the exact same gross amount within 90 days.
- Computes Jaro-Winkler text similarity on service descriptions (>= 90% match triggers REJECT).
- Automatically exempts valid recurring contract POs.

Split Invoice Check (AP-11) -> HOLD

ACTIVE RULE 4

- Checks invoices from the same supplier within a 14-day window.
- Triggers HOLD if 2+ bills each sit within 5% below an approval threshold (INR 50k, 1L, 5L, 10L) and their combined total is >= INR 1,00,000.
- 10 Additional Rules (AP-01, AP-04..08, AP-10, AP-12..14) stored & toggleable via ENABLE_ALL_RULES=true.

6. User-Facing Status Terms & Standardized Output

Plain-language status terms and consistent line-by-line check formatting

6 Plain-Language Display Status Terms

DISPLAY TERMS

PASS

- Passed all checks. Can be saved to the ledger (auto-saved on check).

HOLD

- Needs a review. Not saved to the ledger until cleared in the Review Queue.

REJECT

- Duplicate or split pattern found. Not saved to the ledger until reviewed.

INCOMPLETE

- Required details (Invoice No, Date, Total, or Supplier) are missing.

NOT AN INVOICE

- This is not a supplier invoice (e.g., PO, statement, delivery note).

COULD NOT READ

- The file could not be read (triggers opt-in Extract with AI button).

Fixed 11-Line Check Output Template

CHAT & UI FORMAT

- Every invoice check displays a clean, scannable structure with one item per line and zero internal rule codes:

Example Output:

- Invoice 51109310 -- REJECT
- Reason: This invoice is already saved in the ledger. Uploading it again is treated as a duplicate.
- Duplicate check: failed -- This invoice is already saved in the ledger...
- GSTIN check: passed
- Repeat bill check: passed
- Split invoice check: passed
- Invoice number: 51109310
- Date: 2023-07-03
- Total: INR 18,44,673.60
- Taxable amount: INR 15,63,282.71
- Tax: CGST INR 1,40,695.44, SGST INR 1,40,695.44; IGST not shown on invoice

7. High-Volume Batch Processing & Streaming Exports

Handles up to 2,000 invoices per batch with non-blocking chunked execution

Chunked Background Job Queue (srv/lib/queu

2,000 INVOICES / JOB

- Accepts Fiori CSV exports, Excel (.xlsx) files, or multi-file PDF/image batches up to 2,000 invoices.
- Processes in chunks of 50 invoices per tick via setImmediate so HTTP requests never time out.
- Pre-loads and indexes ledger records in memory (O(1) hash map lookup by GSTIN + Fiscal Year + Invoice No).
- Supports real-time progress polling (GET /jobs/:id), Cancel, Resume, and crash recovery.

Server-Side Pagination & Filtering (pagination.js)

50 ROWS / PAGE

- All tables (S/4HANA Results, Ledger, Review Queue, Check History, Activity Log) use server-side pagination (50 rows per page).
- Instant filtering by Status (All, PASS, HOLD, REJECT) and search by Invoice Number, Supplier Name, or GSTIN.
- Top Summary Bar keeps live synchronized counts: Uploaded, Saved to ledger, Passed, Held, Rejected, Failed to read.

Streaming CSV & Excel (.xlsx) Downloads (srv/lib/export.js)

EXPORTS

Separate One-Click CSV and Excel (.xlsx) Downloads Across Views

- Streams full filtered datasets directly to the browser using ExcelJS workbook writer and RFC-4180 CSV escaping:
- 1. S/4HANA Extract Results (/fraud/exportS4Csv & /fraud/exportS4Excel)
- 2. Central Ledger (/fraud/exportLedgerCsv & /fraud/exportLedgerExcel)
- 3. Check History (/fraud/exportHistoryCsv & /fraud/exportHistoryExcel)

8. Live SAP S/4HANA Cloud Sandbox Integration

Read-only OData V2 connector to SAP API Business Hub in srv/lib/sap.js

SAP API Business Hub OData V2 Integration (API_SUPPLIERINVOICE_PROCESS_SRV)

LIVE READ-ONLY ODATA

Endpoint & Authentication

- Connects to https://sandbox.api.sap.com/s4hanacloud/sap/opu/odata/sap/API_SUPPLIERINVOICE_PROCESS_SRV/A_SupplierInvoice.
- Sends the APIKey header strictly from server-side environment variables (SAP_API_KEY) -- never exposed to the browser.

Strict Read-Only Safety & 5-Second Timeout

- Executes HTTP GET requests only ($top and $filter) with an AbortController 5,000 ms timeout and graceful fallback if SAP_API_KEY is unset.

1. Test SAP Connection

DIAGNOSTIC

- Fetches $top=5 supplier invoices from SAP S/4HANA sandbox.
- Maps SupplierInvoice, FiscalYear, CompanyCode, InvoicingParty, InvoiceGrossAmount, Currency, and Dates.

2. Fetch from SAP Now

BATCH IMPORT

- Pulls live supplier invoices from SAP S/4HANA and feeds them directly into the chunked batch rule engine.
- Tags source as "SAP sandbox (demo data)".

3. Live SAP Cross-Check

LOOKUP

- During single-invoice checks and chat queries, queries SAP by SupplierInvoiceIDByInvcgParty or SupplierInvoice.
- Reports exact match status in the check result.

9. Grounded Read-Only AI Chat Assistant

17-Intent deterministic router + 6 live database tools + Gemini grounding (srv/lib/agent-tools.js)

Step 1 & 2: 17-Intent Router (Pre-LLM)

FAST ROUTING

- Classifies every user message BEFORE calling Gemini:
- - Conversational: GREETING, THANKS_OR_BYE, PERSONAL_OR_OPINION, OFF_TOPIC, COMPLAINT, EMPTY_OR_GIBBERISH, UNCLEAR.
- - Invoice & Rules: INVOICE_DETAILS, INVOICE_RESULT, RULE_QUESTION, BANK_OR_OTHER_FIELD, NEW_CHECK_REQUEST.
- - Data & Security: LEDGER_OR_QUEUE_OR_HISTORY, SYSTEM_OR_CREDENTIALS, RULE_OVERRIDE, ADVICE_REQUEST.
- Guarantees non-invoice messages never leak invoice fields or rule codes.

6 Live Read-Only Database Tools

AGENT TOOLS

- 1. find_by_invoice(invoiceNo) -- Exact & normalized match across Ledger, Review Queue, and Check History.
- 2. find_by_gstin(gstin) -- Lists all posted ledger invoices for a 15-char GSTIN.
- 3. find_by_date(from, to) -- Date & range queries on the ledger.
- 4. find_by_supplier(name) -- Case-insensitive supplier search.
- 5. list_ledger(page) -- Paginated ledger listing (10/page in chat).
- 6. last_check() & get_review_queue() -- Session memory & active HOLD/REJECT queue.

Multi-Lingual Support (English & Telugu) & Session Memory

UX & LANGUAGE

Native Telugu & English Responses

- Detects Telugu script or requests ("explain in Telugu") and replies in clean UTF-8 Telugu without mojibake corruption.

Anti-Repetition Follow-Up Tracker

- Tracks the last follow-up question per user session so the assistant never repeats the exact same closing question twice in a row.

10. Security, Privacy, RBAC & Tamper-Evident Audit Chain

Enterprise guardrails for data privacy, credential protection, and audit compliance

LLM Privacy Guard & Credential Protection

SECURITY

ALLOW_INVOICE_DATA_TO_LLM Guard (Default: false)

- Redacts Supplier Name, GSTIN, Bank IFSC, Account Numbers, Invoice Numbers, and Amounts ([REDACTED_*]) before any prompt is sent to Gemini.

Credential & SQL Injection Refusal

- Any attempt to ask for SAP_API_KEY, GEMINI_API_KEY, server settings, or raw SQL (SELECT/DROP/DELETE) is blocked immediately with a safe refusal.

SHA-256 Tamper-Evident Audit Log (ap.AuditLog)

COMPLIANCE

- Every check, ledger insert, batch job, and alert decision writes an immutable entry to ap.AuditLog.
- Each row stores prevHash and computes:
- hash = SHA256(idx | timestamp | checkId | action | detail | prevHash)
- The Verify Audit Chain endpoint (/fraud/verifyAuditChain) validates the entire cryptographic chain from entry #1.

Role-Based Access Control (RBAC)

3 ROLES

APClerk

- Can upload invoices, run single and batch checks, and post passed invoices to the ledger.

APController

- Can review and decide open HOLD/REJECT alerts in the Review Queue (Approve / Reject / Hold with mandatory audit comment).

Auditor

- Strictly read-only access to view the Ledger, Check History, and Activity Log; blocked (HTTP 403) from checking, importing, posting, or deciding alerts.

11. Deployment Architecture & Complete Technology Stack

Ready for SAP BTP Cloud Foundry (MTA) and local/cloud container environments

Technology Stack Summary

STACK

- Runtime & Framework: Node.js 22, @sap/cds v10, Express 4
- Database: @cap-js/sqlite (in-memory/local) & SAP HANA HDI (@sap/hdi-deploy)
- PDF Text Extraction: pdfjs-dist v4.10.38
- PDF Page Rasterizer: @napi-rs/canvas v1.1.0
- OCR Engine: tesseract.js v7.0.0 + local eng.traineddata
- QR & Crypto: jsqr v1.4.0 + Node.js native RSA-SHA256
- Spreadsheets: exceljs v4.4.0 + custom Fiori CSV parser
- AI SDK: @google/genai v2.32.0 (Gemini 3.8 Flash / 3.5 Flash Lite)

SAP BTP Cloud Foundry Deployment

MTA DEPLOY

- 1. GitHub Actions (.github/workflows/build.yml) runs mbt build -t ./mta_archives on every push to main.
- 2. Packages blus-invoice-agent_1.0.0.mtar containing srv, app, db, data, public, and eng.traineddata.
- 3. Deploy to SAP BTP Cloud Foundry:
- cf deploy mta_archives/blus-invoice-agent_1.0.0.mtar
- 4. Configure live keys via Cloud Foundry env:
- cf set-env blus-invoice-agent-srv GEMINI_API_KEY "..."
- cf set-env blus-invoice-agent-srv SAP_API_KEY "..."

End-to-End Summary

CONCLUSION

- Blus Invoice Agent combines deterministic financial controls (4 active fraud/duplicate rules + arithmetic & GSTIN checksums) with multi-format document extraction (PDF text, Canvas + Tesseract OCR, QR JWT, Fiori CSV/Excel), live SAP S/4HANA OData V2 lookups, and a grounded, read-only AI Assistant.
- The result is zero duplicate payouts, transparent 10-step auditability, and enterprise-grade security on SAP BTP.